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A penalty guided stochastic fractal search approach for system reliability optimization

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ABSTRACT

Modern industry requires components and systems with high reliability levels. In this paper, we address the system reliability optimization problem. A penalty guided stochastic fractal search approach is developed for solving reliability allocation, redundancy allocation, and reliability–redundancy allocation problems. Numerical results of ten case studies are presented as benchmark problems for highlighting the superiority of the proposed approach compared to others from literature.

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1. Introduction

Nowadays, the need of productively competitive and safe industrial plants pressures the designer to develop very highly reliable systems. In order to optimize the system reliability, three main ways may be followed: increasing the component reliability (reliability allocation), using redundant components in parallel (redundancy allocation), and both of them (reliability–redundancy allocation) [1–3]. These ways aim to maximize the system reliability under the considered constraints, such as the system cost, weight and volume [4,5]. The reliability–redundancy allocation is one of the most complex optimization problems, the goal brings to determine the optimal number of redundant components and their reliabilities while respecting the constraints.

Several reliability optimization problems (ROPs) models have been proposed and solved for various types of configurations of the systems. In [1,6–9], a reliability–redundancy allocation model has been investigated for five subsystems connected in series by assuming that the component must not fail in a specified operating time called mission time. A series–parallel connection has been considered under the same assumption of mission time in order to model some replicated subsystems [6,7,9–12]. A mathematical model for an overspeed protection in a gas turbine comprising four subsystems has been considered in order to maximize its overall reliability [1,8–11,13,14]. The reliability optimization problem of various complex configurations has also been investigated, such as complex bridge system [6], structure system [15], and pharmaceutical plant [16]. In most of the above mathematical models, artificial intelligence methods are used as optimizer solution (see Table 1).

The aim of the present work is to develop a ROP solution approach based on a recently invented artificial intelligence method called stochastic fractal search (SFS) [39]. The approach implements a penalty function for solving the ROPs and is called penalty guided stochastic fractal search (PSFS). The remainder of the paper is organized as follows: Section 2 describes the different kinds of ROPs considered; Section 3 introduces the main ideas behind the stochastic fractal search and the penalty approach adopted; in Section 4, ten different numerical case studies are presented in order to highlight the applicability of the proposed approach; Section 6, provides a discussion on the results obtained and a comparison to those available in the literature; finally, some conclusions and suggestions for future work are given in the last section.

2. Problem formulation

As discussed in the previous section, three kinds of reliability optimization problems are considered.

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Notations

R_s	system reliability.
m	number of subsystems in the system.
r_i	reliability of each component in subsystem i, $1 \le i \le m$.
r	$=(r_1, r_2,\ldots, r_m)$, vector of component reliabilities for the system.
n_i	number of components in subsystem i, $1 \le i \le m$.
n	$=(n_1, n_2,\ldots, n_m)$, vector of redundancy allocation for the system.
R_i	$=1-(1-r_i)^{n_i}$, reliability of the ith subsystem, $1 \le i \le m$.
Q_i	$=1-R_i$ unreliability (failure probability) of the ith subsystem, $1 \le i \le m$.
$n_{i,\ \max}$	maximum number of components in subsystem i, $1 \le i \le m$.
M	number of constraints.
g_j	jth constraint function, $j=1,\ldots, M$.
$C(r_i)$	$=\alpha_i(-T/\ln r_i)^{\beta_i}$, cost of each component with reliability r_i at subsystem i, $1 \le i \le m$.
T	operating time during which the component must not fail (mission time).
w_i	weight of each component in subsystem i, $1 \le i \le m$.
v_i	volume of each component in subsystem i, $1 \le i \le m$.
β_i, α_i	parameters representing physical features (shaping and scaling factors, respectively) of each component at subsystem i, $1 \le i \le m$.
W, C	upper limits on the weight and cost of the system, respectively.
V	upper limit on the sum of the subsystems' products of volume and weight.
$\exp(n_i/4)$	term accounting the interconnecting hardware.

2.1. Reliability allocation problem

The general mathematical formulation of the reliability allocation problem is [26,40]:

$$\text{Maximize} \quad R_s(r) = R_s(r_1, r_2, \ldots, r_m) \tag{1}$$

Subject to

$$g_j(r_1, r_2, \ldots, r_m) \le b$$
$$0 \le r_i \le 1; \quad i = 1, 2, \ldots, m$$
$$r_i \in [0, 1] \subset \mathbb{R}^+ \tag{2}$$

where $R_s(\cdot)$ is the objective function of the problem (overall system reliability), $g(\cdot)$ is the set of constraints, such as system weight, volume and cost. r_i is the reliability of the ith subsystem, m is the number of subsystems in the system and b is the vector of resource limitation.

2.2. Redundancy allocation problem

The general mathematical formulation of the redundancy allocation problem is [26,41]:

$$\text{Maximize} \quad R_s(n) = R_s(n_1, n_2, \ldots, n_m) \tag{3}$$

Subject to

$$g_j(n_1, n_2, \ldots, n_m) \le b$$
$$1 \le n_i \le n_{i,\ \max}, \quad i = 1, 2, \ldots, m;$$
$$n_i \in \mathbf{Z}^+ \tag{4}$$

where $R_s(\cdot)$ is the objective function of the problem (overall system reliability), $g(\cdot)$ is the set of constraints, such as system weight, volume and cost, n_i is the number of redundant components in the ith subsystem. m is the number of subsystems in the system and b is the vector of resource limitation.

2.3. Reliability–redundancy allocation problem

The general mathematical formulation of the reliability–redundancy allocation problem is [7,11,26]:

$$\text{Maximize} \quad R_s(r, n) = R_s(r_1, r_2, \ldots, r_m;\ n_1, n_2, \ldots, n_m) \tag{5}$$

Subject to

$$g_j(r_1, r_2, \ldots, r_m;\ n_1, n_2, \ldots, n_m) \le b$$
$$0 \le r_i \le 1; \quad 1 \le n_i \le n_{i,\ \max}, \quad i = 1, 2, \ldots, m;$$
$$r_i \in [0, 1] \subset \mathbb{R}^+; \quad n_i \in \mathbf{Z}^+ \tag{6}$$

where $R_s(\cdot)$ is the objective function of the problem (overall system reliability), $g(\cdot)$ is the set of constraints, such as system weight, volume and cost. r_i and n_i are the reliability and the number of redundant components in the ith subsystem, respectively, m is the number of subsystems in the system and b is the vector of resource limitation. It is noted that the RRAP involves integer and real decision variables at the same time.

3. Penalty guided stochastic fractal search

The stochastic fractal search (SFS) is an evolutionary algorithm proposed by Salimi [39] for solving optimization problems. The basic concept of SFS is founded on the property of an entity (object or quantity) for exploring an area. The fractal theories define the mathematical model of diffusing a particle and self-similarity of the patterns in nature, such as trees, snowflakes, animal coloration patterns, crystals, Romanesco broccoli, lungs, river networks, blood vessels and DNA. The SFS dealt with attractable problems, such as: optimizer solver in precise trajectory optimization [42], geometric transformations [43], and distributed database queries [44].

The basic principles of the SFS are as follows [39]:

- Initialize a random sample of point cloud solutions.
- Move the points using a diffusion process (random walk).
- Update the position of each point according to the quality of the solution and identify the best one.
- Repeat the procedures until the stopping criterion is satisfied.

In this paper, the stochastic fractal search proposed in [39] is implemented with an original penalty guided function for solving the three above reliability optimization problems described in Section 2. The overall baseline of the PSFS is:

Step 1: Input the parameters.

Step 2: Generate the initial vector of λ solution points. Each solution point is a vector of number of components n_i or/and of their reliabilities r_i, $i=1,2,\ldots,m$.

Step 3: Evaluate the objective function.

For evaluating the overall system reliability, the fitness function is written as follows:

$$\text{Objective function} = -R_s(r_1, r_2, \ldots, r_m, n_1, n_2, \ldots, n_m) \tag{7}$$

The sign $(-)$ is conveniently introduced for maximizing the function.

Step 4: Constraint handling.

Since the reliability optimization problems contain nonlinear constraint functions (such as limits on cost, weight, and volume), a

handling method to guide the search towards the best solutions within the feasible space is needed. The set of variables affected to the redundancy allocation (n_1, n_2, …, n_m) is rounded to the nearest integer value. Most of the constraint handling methods can be listed according to their fundament into five categories [1]: (i) penalty functions, (ii) preserving feasibility of solutions, (iii) decoders, (iv) making a distinction between feasible and infeasible solutions, and (v) hybrid methods.

In this paper, an original penalty function is implemented in order to eliminate the infeasible solutions (i.e. solutions leading to constraint violation). The main idea is to add penalty terms to the objective function value [45]. Here, it is applied as follows:

$$\text{Fitness value} = -R_s(r_1, r_2, ..., r_m, n_1, n_2, ..., n_m) + \psi(r_1, r_2, ..., r_m, n_1, n_2, ..., n_m) \tag{8}$$

where $\psi(r_1, r_2, \ldots, r_m, n_1, n_2, \ldots, n_m)$ is the penalty function, calculated as follows:

$$\psi(r_1, r_2, ..., r_m, n_1, n_2, ..., n_m) = \sum_{j=1}^{M} \phi_j \cdot \max(0, g_j(r_1, r_2, ..., r_m, n_1, n_2, ..., n_m))^2 \tag{9}$$

where ϕ_j is a positive constant called the penalty parameter or penalty factor. If the operand of the penalty term is negative, then, the operand takes the absolute value; otherwise, it is set to zero. In Eq. (8), the penalty term is added in order to decrease the fitness of the infeasible solutions and pressure the selection of the feasible solutions.

Step 5: Diffusion process (search for new solutions).

The new solutions are created by diffusing new solution points according to Gaussian random walks which represent a probability as follows:

$$GW = Gaussian(\mu_p, \sigma) = \sqrt{2\tau/(\pi\sigma)}e^{-(\mu_p/\delta)^2\tau/\sigma} \tag{10}$$

where μ_p is a parameter equal to the absolute value of the ith solution point in the group denoted $|P_i|$ and σ is the standard deviation, which is calculated as follows: $\sigma = \left|\frac{\log(N_{Iter})}{N_{Iter}} \times (P_i - BP)\right|$, where N_{Iter} is the number of iterations and BP is the position of the best solution point. τ is the jump length and δ is the time between two successive jumps.

Step 6: Two updating processes.

This step is performed in order to undertake the intensification process of the evolutionary algorithm and preserve good solutions for the next iteration.

For the first updating process, the solution points are ranked based on their respective fitness values as follows:

$$P_{a_i} = \frac{rank(P_i)}{\lambda} \tag{11}$$

where P_{a_i} is a probability assumed that its highest value is affected to the best point, $rank(P_i)$ is the rank of the point P_i in the group and λ is the number of points.

A random number ε is generated in [0, 1] for each point P_i. If $P_{a_i} < \varepsilon$, then, the kth component of P_i is updated as follows:

$$P_i'(k) = P_r(k) - \varepsilon \times \left[(P_t(k) - P_i(k))\right] \tag{12}$$

where P_i' is the new position, P_r and P_t are solution points selected randomly in the group.

For the second updating process, the solution points obtained in the first updating process are ranked according to Eq. (11). The position of P_i' is modified again as follows:

$$P_i'' = P_i' - \hat{\epsilon} \times (P_t' - BP) \quad \text{if } \epsilon' \le 0.5 \tag{13}$$

$$P_i'' = P_i' + \hat{\epsilon} \times (P_t' - P_r') \quad \text{if } \epsilon' > 0.5 \tag{14}$$

Table 1
Methods applied in the literature.

Abbreviation	Method
GA [7,17]	Genetic algorithms.
CS–GA [18]	Hybrid cuckoo search and genetic algorithm.
GA–SRS [19]	Genetic algorithm with standby redundancy strategy.
ES–PSO [20]	Eagle strategy with particle swarm optimization.
ICS [14]	Improved cuckoo search.
ABC [8,21]	Artificial bee colony.
BBO [1]	Biogeography based optimization algorithm.
CS [9,12]	Cuckoo search.
MPSO [22]	Modified particle swarm optimization.
IPSO [23]	Improved particle swarm optimization.
INGHS [24]	Improved novel global harmony search.
IA [11]	Immune based algorithm.
IAs [10]	Artificial immune algorithm.
LXPM–IPSO–GS [25]	Hybrid approach based on Laplace crossover power mutation genetic algorithm, improved particle swarm optimization, and golden search.
HNN–SC [6]	Surrogate-constraints algorithm.
XKL [26]	Analytic algorithms and local strategy.
AR–ICA [27]	Imperialist competitive algorithm with attraction and repulsion concepts.
SSO [28]	Simplified swarm optimization.
PSSO [28]	Particle-based simplified swarm optimization.
GA–PSO [29]	Hybrid genetic algorithm–particle swarm optimization.
DE [30]	Differential evolution with Lévy flight.
TS–DE [31]	Hybrid Tabu search–differential evolution algorithm.
PSO–GC [13]	Particle swarm optimization based on Gaussian distribution and chaotic sequence.
NMDE [32]	Novel modified differential evolution algorithm.
FGO [33]	Fuzzy global optimization.
I–NESA [34]	Improved nonequilibrium simulated annealing.
C–SOMGA [35]	Self organizing migrating genetic algorithm.
MRPS [36]	Modified random-to-pattern search.
SA [34]	Simulated annealing.
LXPM–IPSO [25]	Laplace crossover power mutation genetic algorithm with improved particle swarm optimization.
HS–IPSO–GA [25]	Hybrid approach based on harmony search, improved particle swarm optimization, and genetic algorithm.
HS [12]	Harmony search.
NGHS [24]	Novel global harmony search.
AGGA [37]	Genetic search.
ASACO [37]	Adaptive penalty function with the basic ant colony approach.
NAFSA [38]	Novel artificial fish swarm algorithm

where P_r' and P_t' are random points selected from the first procedure, $\hat{\varepsilon}$ is a random number generated by the Gaussian Normal distribution for each index i, and ε' is randomly generated in [0, 1] for each index i.

Step 7: If the number of iterations is reached, stop and display the optimal results; otherwise, go to **Step 3**.

The pseudo-code of the penalty guided stochastic fractal search implemented for solving reliability optimization problems is presented in Algorithm 1, whereas its general flowchart is shown in Fig. 1.

Algorithm 1. Pseudo-code of the implemented PSFS.

1: Input the parameters: λ, N_{Iter}, ϕ_j.
2: Generate an initial vector of λ solution points.
3: While $z < N_{Iter}$
4: Evaluate the fitness function according to Eq. (7).
5: Constraint handling according to Eqs. (8) and (9).
6: Diffusion process according to Eq. (10).
7: First updating process performed by Eqs. (11) and (12).
8: Second updating process performed by Eqs. (11), (13), and (14).
9: End while
10: Display the results.

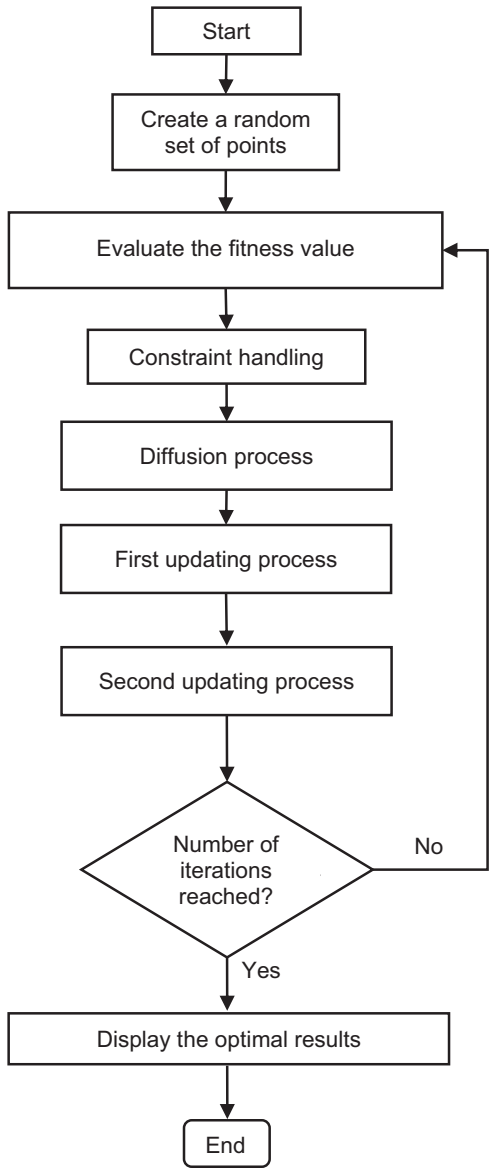

Fig. 1. Flowchart of the implemented PSFS.

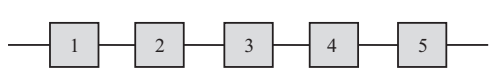

Fig. 2. Series system.

4. Case studies

This section presents ten case studies with different kinds of system reliability optimization problems.

4.1. Series system

The series system [1,6–9] considered in this case study connects five subsystems, as shown in Fig. 2. The mathematical model is a nonlinear mixed-integer programming problem involving ten decision variables, five real variables for the reliability allocation and five integer variables for the redundancy allocation (Table 2):

$$\text{Maximize} \quad R_s(r,n) = \prod_{i=1}^{5} \left[1-(1-r_i)^{n_i}\right] \tag{15}$$

Table 2
Data used in series system and complex-bridge-system.

Subsystem i	$10^5\alpha_i$	β_i	v_i	w_i	V	C	W	T (h)
1	2.330	1.5	1	7	110	175	200	1000
2	1.450	1.5	2	8				
3	0.541	1.5	3	8				
4	8.050	1.5	4	6				
5	1.950	1.5	2	9				

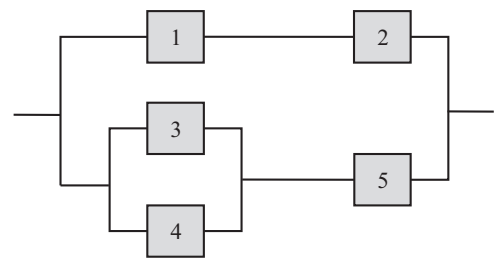

Fig. 3. Series–parallel system.

Subject to

$$g_1(r,n) = \sum_{i=1}^{5} v_i n_i^2 \le V \tag{16}$$

$$g_2(r,n) = \sum_{i=1}^{5} C(r_i)\left[n_i + \exp(n_i/4)\right] \le C \tag{17}$$

$$g_3(r,n) = \sum_{i=1}^{5} w_i n_i \exp(n_i/4) \le W \tag{18}$$

$$0.5 \le r_i \le 1,\ r_i \in [0,1] \subset \mathbb{R}^+;\ 1 \le n_i \le 5,\ n_i \in \mathbf{Z}^+; i = 1,2,\ldots,5$$

4.2. Series–parallel system

Fig. 3 shows the series–parallel system considered. The mathematical model is given as follows [6,7,9,10–12] (Table 3):

$$\text{Maximize} \quad R_s(r,n) = 1-(1-R_1R_2)[1-(R_3+R_4-R_3R_4)R_5] \tag{19}$$

Subject to
$g_1(r,n)$, $g_2(r,n)$, $g_3(r,n)$ (as specified by Eqs. (16)–(18)).

$$0.5 \le r_i \le 1,\ r_i \in [0,1] \subset \mathbb{R}^+; 1 \le n_i \le 5,\ n_i \in \mathbf{Z}^+;\ i = 1,2,\ldots,5$$

4.3. Complex (bridge) system

The overall reliability of the bridge system (see Fig. 4) is given as follows [1,6–10,13,14]:

$$\text{Maximize} \quad R_s(r,n) = R_5(1-Q_1Q_3)(1-Q_2Q_4) + Q_5[1-(1-R_1R_2)(1-R_3R_4)] \tag{20}$$

Subject to
$g_1(r,n)$, $g_2(r,n)$, $g_3(r,n)$ (as specified by Eqs. (16)–(18))

$$0.5 \le r_i \le 1,\ r_i \in [0,1] \subset \mathbb{R}^+; \quad 1 \le n_i \le 5,\ n_i \in \mathbf{Z}^+; \quad i = 1,2,\ldots,5$$

4.4. Overspeed protection system

The reliability–redundancy allocation problem of the overspeed protection system for a gas turbine (see Fig. 5) is formulated as

Table 3
Data used in series–parallel system.

Subsystem i	$10^5\alpha_i$	β_i	v_i	w_i	V	C	W	T (h)
1	2.500	1.5	2	3.5	180	175	100	1000
2	1.450	1.5	4	4				
3	0.541	1.5	5	4				
4	0.541	1.5	8	3.5				
5	2.100	1.5	4	4.5				

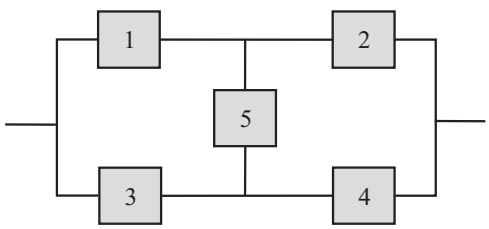

Fig. 4. Complex (bridge) system.

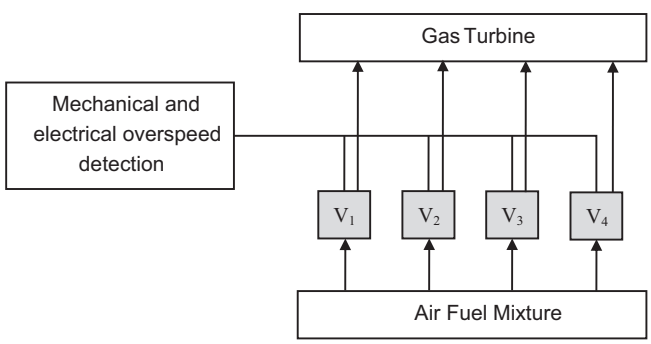

Fig. 5. Overspeed protection system.

follows [1,8–11,13,14]:

$$\text{Maximize} \quad R_s(r,n) = \prod_{i=1}^{4} \left[1-(1-r_i)^{n_i}\right] \tag{21}$$

Subject to

$$g_1(r,n) = \sum_{i=1}^{4} v_i n_i^2 \le V \tag{22}$$

$$g_2(r,n) = \sum_{i=1}^{4} \alpha_i(-T/\ln r_i)^{\beta_i}\left[n_i + \exp(n_i/4)\right] \le C \tag{23}$$

$$g_3(r,n) = \sum_{i=1}^{4} w_i n_i \exp(n_i/4) \le W \tag{24}$$

$$0.5 \le r_i \le 1,\ r_i \in [0,1] \subset \mathbb{R}^+; \quad 1 \le n_i \le 10,\ n_i \in \mathbf{Z}^+; \quad i = 1,2,\ldots,4$$

The problem involves four valves (V_1–V_4) and the goal is to determine the optimal r_i and n_i values (eight decision variables), i=1, 2,…,4. These valves must close to cut off the fuel supply when an overspeed occurs. The overspeed detection is continuously provided by the electrical and mechanical systems (Table 4).

4.5. Complex bridge network system

The complex bridge network system [8,33,34] includes five subsystems, as shown in Fig. 6. The goal is to minimize the cost by considering the reliability constraint. The mathematical problem

Table 4
Data used in overspeed protection system.

Subsystem i	$10^5\alpha_i$	β_i	v_i	w_i	V	C	W	T (h)
1	1.0	1.5	1	6	250	400	500	1000
2	2.3	1.5	2	6				
3	0.3	1.5	3	8				
4	2.3	1.5	2	7				

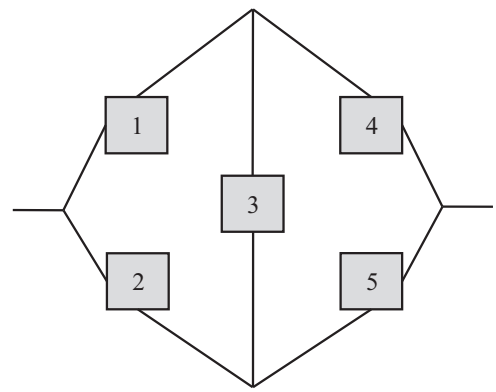

Fig. 6. Complex bridge network system.

of the system is defined as follows:

$$R_s(r_1,r_2,r_3,r_4,r_5) = r_1r_4 + r_2r_5 + r_2r_3r_4 + r_1r_3r_5 + 2r_1r_2r_3r_4r_5 - r_2r_3r_4r_5 - r_1r_3r_4r_5 - r_1r_2r_4r_5 - r_1r_2r_3r_5 - r_1r_2r_3r_4 \tag{25}$$

$$C_i = a_i \exp\left(\frac{b_i}{1-r_i}\right) \tag{26}$$

$$C_s = \sum_{i=1}^{5} C_i \tag{27}$$

$$\text{Minimize} \quad C_s = \sum_{i=1}^{5} a_i \exp\left(\frac{b_i}{1-r_i}\right) \tag{28}$$

Subject to

$$0.5 \le r_i \le 1, \quad \text{for } i = 1,2,3,4,5 \tag{29}$$

$$0.99 \le R_s \le 1 \tag{30}$$

$$a_i = 1; \quad b_i = 0.0003, \quad \forall i$$

4.6. Life-support system in a space capsule

As for the complex bridge network, the goal in the life-support system [33,34] is to minimize the cost under the system reliability constraint:

$$R_s(r_1,r_2,r_3,r_4) = 1 - r_3[(1-r_1)(1-r_4)]^2 - (1-r_3)[1-r_2(1-(1-r_1)(1-r_4))]^2 \tag{31}$$

$$\text{Minimize} \quad C_s = \sum_{i=1}^{4} K_i\left[\tan\left(\frac{\pi}{2}r_i\right)\right]^{\alpha_i} \tag{32}$$

$$0.5 \le r_i \le 1, \quad \text{for } i = 1,2,3,4 \tag{33}$$

$$0.99 \le R_s \le 1 \tag{34}$$

$\alpha_i = 1, \ \forall i$ where K_1=K_2=25, K_3=50, and K_4=37.5.

Table 5
Data used in large scale reliability–redundancy allocation problem.

Subsystem i	$10^5\alpha_i$	β_i	v_i	w_i	V	C	W	T (h)
1	0.6	1.5	2	8	600	700	900	1000
2	0.1	1.5	5	9				
3	1.2	1.5	5	6				
4	0.3	1.5	4	10				
5	2.9	1.5	4	8				
6	1.7	1.5	1	9				
7	2.6	1.5	1	9				
8	2.5	1.5	4	7				
9	1.3	1.5	4	9				
10	1.8	1.5	3	8				
11	2.4	1.5	3	9				
12	1.3	1.5	1	8				
13	1.2	1.5	1	7				
14	2.1	1.5	3	10				
15	0.9	1.5	4	6				
16	1.3	1.5	5	7				
17	1.9	1.5	1	7				
18	2.7	1.5	4	8				
19	2.8	1.5	2	9				
20	1.5	1.5	1	9				

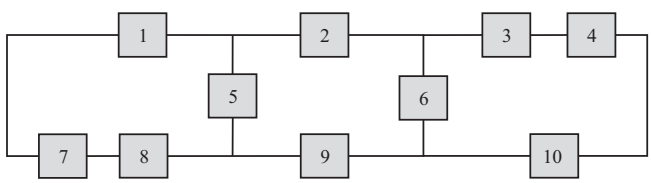

Fig. 7. 10-unit structure system.

4.7. Large scale reliability–redundancy allocation problem

The proposed approach has been applied on a large scale reliability–redundancy allocation problem involving forty decisions variables [25] (Table 5):

$$\text{Maximize} \quad R_s(r,n) = \prod_{i=1}^{20} \left[1-(1-r_i)^{n_i}\right] \tag{35}$$

Subject to

$$g_1(r,n) = \sum_{i=1}^{20} v_i n_i^2 \le V \tag{36}$$

$$g_2(r,n) = \sum_{i=1}^{20} \alpha_i(-T/\ln r_i)^{\beta_i}\left[n_i + \exp(n_i/4)\right] \le C \tag{37}$$

$$g_3(r,n) = \sum_{i=1}^{20} w_i n_i \exp(n_i/4) \le W \tag{38}$$

$$0.5 \le r_i \le 1, \quad r_i \in [0,1] \subset \mathbb{R}^+; \quad 1 \le n_i \le 10, \quad n_i \in \mathbf{Z}^+; \quad i = 1,2,\ldots,20$$

4.8. 10-unit structure system

The 10-unit structure system (see Fig. 7) is defined as follows [12,15,37]:

$$\begin{aligned}\text{Maximize } R_s(n) = {} & R_1R_2R_3R_4 + R_1R_2R_6R_{10}(Q_3+R_3Q_4) + R_1R_5R_9R_{10} \\ & \times(Q_2+R_2Q_3Q_6+R_2R_3Q_4Q_6) + R_7R_8R_9R_{10} \\ & \times(Q_1+R_1Q_2Q_5+R_1R_2Q_3Q_5Q_6+R_1R_2R_3Q_5Q_6Q_4) \\ & +(R_2R_3R_4R_5R_7R_8Q_1(Q_9+R_9Q_{10}) \\ & +Q_1R_3R_4R_6R_7R_8R_9Q_{10}(Q_2+R_2Q_5)+R_1Q_2R_3R_4R_6R_7R_8R_9Q_{10} \\ & +R_1Q_2R_3R_4R_5R_6R_9Q_{10}(Q_7+R_7Q_8) \\ & +Q_1R_2R_5R_6R_7R_8Q_9R_{10}(Q_3+R_3Q_4)\end{aligned} \tag{39}$$

Table 6
Data used in 10-unit and 15-unit structure systems.

Subsystem i	r_i	C_1	C_2	C_3	C_4	C_5
1	0.6796	33.2468	35.6054	13.7848	44.1345	10.9891
2	0.7329	27.5668	44.9520	96.7365	25.9855	68.0713
3	0.6688	13.3800	28.6889	85.8783	19.2621	1.0164
4	0.6102	0.4710	0.4922	63.0815	12.1687	29.4809
5	0.7911	51.2555	39.6833	78.5364	23.9668	59.5441
6	0.8140	82.9415	59.2294	11.8123	28.9889	46.5904
7	0.8088	51.8804	78.4996	97.1872	47.8387	49.6226
8	0.7142	77.9446	86.6633	45.0850	25.0545	59.2594
9	0.8487	26.8835	7.8195	3.6722	76.9923	87.4070
10	0.7901	85.8722	27.7460	55.3950	53.3007	55.3175
11	0.6972	41.8733	90.4377	75.7999	95.0057	54.1269
12	0.6262	61.6181	58.0131	98.5166	97.9127	59.1341
13	0.6314	90.0418	77.8206	60.6308	37.2226	40.9427
14	0.6941	75.5947	36.4524	70.4654	96.9179	40.2141
15	0.6010	88.5974	61.0591	18.8802	42.1222	80.0045
d	–	3.2150	3.4710	3.3247	2.6236	3.4288

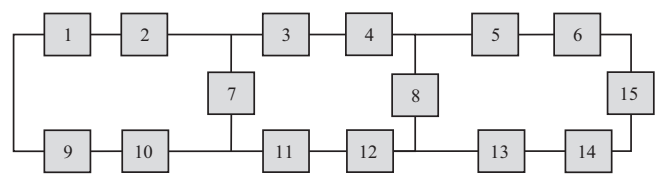

Fig. 8. 15-unit structure system.

Subject to

$$g_y(n) = \sum_{i=1}^{10} c_{yi} n_i \le b_y, \quad y = 1,2,\ldots,M \tag{40}$$

$$n_i \ge 1, \quad n_i \in \mathbf{Z}^+; \quad i = 1,2,\ldots,10$$

where $b_y = d \times \sum_{i=1}^{10} c_{yi}$ with d=rand(1.5,3.5). The coefficients c_{yi} and r_i are generated from uniform distributions in [0, 100] and [0.6, 0.85], respectively. M refers to the number of constraints (two scenarios are considered, M=1 and M=5) (Table 6).

4.9. 15-unit structure system

The system shown in Fig. 8 contains four additional subsystems in comparison to the 10-unit structure system. The system reliability problem is expressed as [9,14,37]:

$$\begin{aligned}\text{Maximize } R_s(n) = {} & R_1R_2R_3R_4R_5R_6 + R_9R_{10}R_{11}R_{12}R_{13}R_{14}R_{15} \\ & \times(Q_1+R_1Q_2+R_1R_2Q_3+R_1R_2R_3Q_4+R_1R_2R_3R_4Q_5+R_1R_2R_3R_4R_5Q_6) \\ & +R_3R_4R_5R_6R_7R_9R_{10}(Q_{11}+R_{11}Q_{12}+R_{11}R_{12}Q_{13} \\ & +R_{11}R_{12}R_{13}Q_{14}+R_{11}R_{12}R_{13}R_{14}Q_{15}) \times (Q_1+R_1Q_2) \\ & +((Q_1+R_1Q_2)(Q_3+R_3Q_4+R_3R_4Q_7)+R_1R_2Q_7(Q_3+R_3Q_4)) \\ & \times(Q_{13}+R_{13}Q_{14}+R_{13}R_{14}Q_{15})R_5R_6R_8R_9R_{10}R_{11}R_{12} \\ & +R_1R_2R_5R_6R_7R_8R_{11}R_{12}(R_9R_{10}+Q_9+R_9Q_{10}) \\ & \times(Q_3+R_3Q_4)(Q_{13}+R_{13}Q_{14}+R_{13}R_{14}Q_{15}) \\ & +(Q_5+R_5Q_6) \times ((Q_7+R_7Q_{11}+R_7R_{11}Q_{12})(Q_9+R_9Q_{10}) \\ & +R_9R_{10}(Q_{11}+R_{11}Q_{12})) \times R_1R_2R_3R_4R_8R_{13}R_{14}R_{15} \\ & +R_1R_2R_7R_{11}R_{12}R_{13}R_{14}R_{15}(Q_9+R_9Q_{10})(Q_3+R_3Q_4+R_3R_4Q_5+R_3R_4R_5Q_6) \\ & +R_3R_4R_7R_8R_9R_{10}R_{13}R_{14}R_{15}(Q1+R_1Q_2) \\ & \times(Q_{11}+R_{11}Q_{12})(Q_5+R_5Q_6)\end{aligned} \tag{41}$$

Subject to

$$g_y(n) = \sum_{i=1}^{15} c_{yi} n_i \le b_y, \quad y = 1,2,\ldots,M \tag{42}$$

$n_i \geq 1,\ n_i \in \mathbf{Z}^+;\quad i = 1, 2, ..., 15$ where the parameters $R_i = 1-(1-r_i)^{n_i}$, $Q_i = 1-R_i = (1-r_i)^{n_i}$, c_{yi}, r_i, and M are similar to those used in the 10-unit structure, and $b_y = rand(1.5, 3.5) \times \sum_{i=1}^{15} c_{yi}$.

4.10. Pharmaceutical plant

The pharmaceutical plant (see Fig. 9) described by Garg and Sharma [16] contains ten subsystems connected in series. The raw material chronologically follows the production line: weighting machine, sifter machine, mass mixer, granulator, fluid bed dryer, octagonal blender, rotary compression machine, coating machine, air compressor and finally the strip packing machine.

The overall system reliability problem of the pharmaceutical plant is given as follows (Table 7):

$$\text{Maximize} \quad R_s(r,n) = \prod_{i=1}^{10} \left[1-(1-r_i)^{n_i}\right] \tag{43}$$

Subject to

$$g_2(r,n) = \sum_{i=1}^{10} \alpha_i(-T/\ln r_i)^{\beta_i}\left[n_i + \exp(n_i/4)\right] \leq C \tag{44}$$

$$g_3(r,n) = \sum_{i=1}^{10} w_i n_i \exp(n_i/4) \leq W \tag{45}$$

$$g_1(r,n) = \sum_{i=1}^{10} v_i n_i^2 \leq V \tag{46}$$

$$0.5 \leq r_i \leq 1-10^{-6},\ r_i \in \left[0, 1-10^{-6}\right] \subset \mathbb{R}^+;\ 1 \leq n_i \leq 5,\ n_i \in \mathbf{Z}^+;\ i = 1, 2, ..., 10$$

5. Results

The implemented PSFS has been programmed using Matlab and run on an Intel Pentium Processor G620 with 4 GB of RAM and (3Mo Cache, 2.60 GHz, Sandy Bridge, Windows 7, 64 bits). The number of function evaluations (NFE) is 5000 for all the case studies. It has been fixed based on trial-and-error method. The performance of the proposed approach is compared to other methods applied in the literature by calculating the maximum possible improvement (MPI) as follows [1,9,13,32]:

$$\text{MPI}(\%) = 100\% \times \left(\frac{R_s^{PSFS} - R_s^{Other}}{1 - R_s^{Other}}\right) \tag{47}$$

where R_s^{PSFS} is the solution of obtained by the penalty guided stochastic fractal search and R_s^{Other} is the solution obtained by another method applied in the literature. Furthermore, 50 independent runs have been considered for each case study in order to provide the standard deviation (SD) of the PSFS.

The numerical results for all the presented case studies are reported in Tables 8–17, where the best system reliability value, number of function evaluations and standard deviation are highlighted in bold type. Furthermore, each Table compares the results obtained by the implemented PSFS to other methods applied in the literature. In system reliability, it should be noted that the smallest improvement is crucial and welcomed for the safety and efficiency of the system function.

6. Discussion

For the series system (see Table 8), the optimal system reliability is 0.931682387907 and reveals an improvement compared to twenty four methods considered for comparison, such as the GA [7], HNN–SC [6], SSO [28], GA–SRS [19], NAFSA [38], CS [9] and PSO [28]. The improvements with respect to these methods were 0.1525%, 0.3381%, 0.2633%, 3.6663%, 0.0001747%, 0.0004117% and 38.7265%, respectively. Two methods have exceeded the constraint limits, namely the CS [12] and ABC [21]. From Table 8, it can be observed that the PSFS provided better system reliability using the lowest number of function evaluations (5000) and the smallest SD (2.5E−16), including the recently published works, i.e. the INGHS [24] (150,000; 4.5609E−03), CS [9] (1.49487E−05), BBO [1] (20,000; 4.00634E−09), PSSO [28] (100,000; 0.026718) and NAFSA [38] (50,000; 0.0067994).

In Table 9, the maximum overall reliability obtained by the PSFS for the series–parallel system is 0.99997664 which is superior to all other methods, such as PSO [28] by 83.5034%, HNN–SC [6] by 25.2770%, GA–SRS [19] by 23.0898%, GA [7] by 9.5626%, SSO [28] by 0.3374%, PSSO [28] by 0.001404%, and NAFSA [38] by 0.004548%. As in the series system, some works provide infeasible solutions, such as the GA–PSO [29], TS–DE [31], and CS–GA [18]. From Table 10, it can be observed that the overall reliability of the complex (bridge) system which is obtained by the PSFS is 0.99988963751 and has outperformed twenty nine methods considered. The best value has been obtained by the NMDE [32] with 0.99988963755, but used 40,000 function evaluations. For both systems, the PSFS is better than the other methods in terms of system reliability, NFE and SD.

The results for the overspeed protection system, complex bridge network system and life-support system in a space capsule are shown in Tables 11–13, respectively. Comparing the results,

Table 7
Data used in pharmaceutical plant.

Subsystem i	$10^5\alpha_i$	β_i	v_i	w_i	V	C	W	T (h)
1	0.611360	1.5	4	9	289	553	483	1000
2	4.032464	1.5	5	7				
3	3.578225	1.5	3	5				
4	3.654303	1.5	2	9				
5	1.163718	1.5	3	9				
6	2.966955	1.5	4	10				
7	2.045865	1.5	1	6				
8	2.649522	1.5	1	5				
9	1.982908	1.5	4	8				
10	3.516724	1.5	4	6				

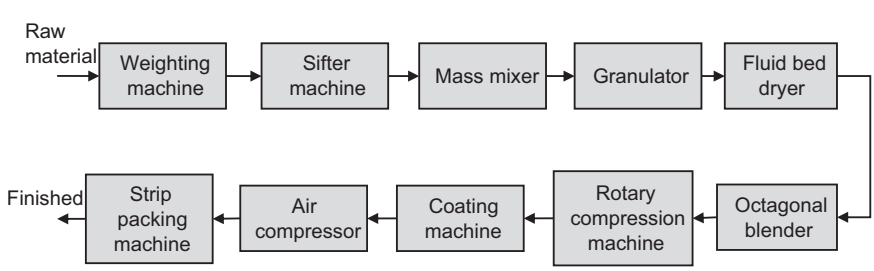

Fig. 9. Pharmaceutical plant.

Table 8
Results for the series system.

Method	(n_1,n_2,n_3,n_4,n_5)	r_1	r_2	r_3	r_4	r_5	R_s	Slack (g_1)	Slack (g_2)	Slack (g_3)	MPI (%)	NFE	SD
HNN–SC [6]	(3,2,2,3,3)	0.77488692	0.87006548	0.89854931	0.71652370	0.79136814	0.93145061	27	0.10824363	7.5189182	0.3381	–	–
GA [7]	(3,2,2,3,3)	0.779427	0.869482	0.902674	0.714038	0.786896	0.931578	27	0.121454	7.518918	0.1525	100,000	–
ES–PSO [20]	(3,2,2,3,3)	0.7793997	0.8718379	0.9028848	0.7114028	0.7877971	0.931682378235[a]	27	$1.64373786 \times 10^{-5}$	7.5189182411	1.4157×10^{-5}	–	3.79E−06
GA–SRS [19]	(3,2,2,3,3)	0.76459335	0.88752892	0.91539527	0.69350544	0.77603145	0.929082263568[b]	27	$2.26949936 \times 10^{-5}$	7.5189182411	3.6663	20,000	5.98581E−04
INGHS [24]	(3,2,2,3,3)	0.7793988710	0.8718370210	0.9028853550	0.7114025151	0.787799488032	0.931682387723	27	3.118×10^{-7}	7.5189182411	2.6933×10^{-7}	150,000	4.5609E−03
CS [12]	(3,2,2,3,3)	0.77941694	0.87183328	0.90288508	0.71139387	0.78780371	0.93168276782[c]	27	$-7.71106726 \times 10^{-7}$	7.5189182411	*	–	
IA [11]	(3,2,2,3,3)	0.779462304	0.871883456	0.902800879	0.711350168	0.787861587	0.931682340	27	5.284×10^{-7}	7.518918	7.0123×10^{-5}	50,000	1.3E−14
IAs [10]	(3,2,2,3,3)	0.779266	0.872513	0.902634	0.710648	0.788406	0.931678	27	1.559×10^{-3}	7.518918	0.006422	13,500	–
LXPM–IPSO–GS [25]	(3,2,2,3,3)	0.779509	0.871859	0.902891	0.711345	0.787739	0.93168236	27	2.20×10^{-7}	7.518918	4.0848×10^{-5}	20,000	2.324E−03
ABC [21]	(3,2,2,3,3)	0.779399	0.871837	0.902885	0.711403	0.787800	0.931682	27	-2.183652×10^{-4}	7.518918241	*	–	–
XKL [26]	(3,2,2,3,3)	0.77939	0.87183	0.90288	0.71139	0.78779	0.93167	27	0.014	7.519	0.01812	–	–
GA [17]	(3,2,2,3,3)	0.780874	0.871292	0.902316	0.711945	0.786995	0.9316734167501[d]	27	0.00395145681	7.5189182411	0.01312	14,000	–
MPSO [22]	(3,2,2,3,3)	0.7793996871	0.8718379458	0.9028848599	0.7114027590	0.7877970932	0.931682387881029[e]	27	$2.2793784637 \times 10^{-8}$	7.518918241159	3.8015×10^{-8}	–	3.84E−05
CS [9]	(3,2,2,3,3)	0.779439734086	0.871995212292	0.902873050171	0.711127088245	0.787986374473	0.931682106582	27	$4.42986674 \times 10^{-7}$	7.518918241	0.0004117	–	1.49487E−05
ICS [14]	(3,2,2,3,3)	0.779416938	0.871833278	0.902885082	0.711393868	0.787803712	0.931682387	27	2.65×10^{-7}	7.518918241	1.3276×10^{-6}	10,000	1.99046E−03
IPSO [23]	(3,2,2,3,3)	0.78037307	0.87178343	0.90240890	0.71147356	0.78738760	0.931680	27	1.01×10^{-4}	7.518918	0.003495	13,500	5.2382E−03
AR–ICA [27]	(3,2,2,3,3)	0.779874	0.872057	0.903426	0.710960	0.786902	0.93167939	27	9.9×10^{-5}	7.518918	0.004387	13,500	0.01863188
ABC [8]	(3,2,2,3,3)	0.779403565208	0.87183320141	0.902886411643	0.711398061305	0.787808548579	0.931682387672	27	2.258957×10^{-10}	7.518918241	3.4398×10^{-7}	20,000	2.37214E−08
BBO [1]	(3,2,2,3,3)	0.7794057271	0.8718339252	0.9028816332	0.7114011594	0.7878058784	0.9316823874	27	4.7045×10^{-7}	7.518918	7.4212×10^{-7}	20,000	4.00634E−09
CS–GA [18]	(3,2,2,3,3)	0.779398871	0.871837021	0.902885355	0.711402515	0.787799488	0.9316823876898[f]	27	$3.6977516515 \times 10^{-7}$	7.5189182411	3.1792×10^{-7}	30,000	8.15E−04
PSO [28]	(2,3,2,4,2)	0.80059281	0.74049316	0.82914384	0.63686144	0.88704276	0.8885037	4	16.775727	4.8146147	38.7265	100,000	–
SSO [28]	(3,2,2,3,3)	0.78271484	0.8735199	0.90264893	0.71313477	0.77729797	0.93150199	27	0.00182140	7.51891824	0.2633	100,000	–
PSSO [28]	(3,2,2,3,3)	0.77946645	0.87173278	0.90284951	0.71148780	0.78781644	0.931682297215271[g]	27	$4.90819664 \times 10^{-5}$	7.51891824	0.0001327	100,000	0.026718
NAFSA [38]	(3,2,2,3,3)	0.77938841387	0.87172098236	0.90303339184	0.71141836221	0.78778928811	0.93168226855	27	6.7347×10^{-9}	7.518918	0.0001747	50,000	0.0067994
PSFS (Proposed approach)	(3,2,2,3,3)	0.77939888253	0.8718370134	0.90288535705	0.71140251717	0.7877994847351	**0.931682387907**	27	$4.132516551 \times 10^{-11}$	7.5189182411	–	**5000**	**2.5E−16**

Slack: the unused resource.
NFE: number of function evaluations.
SD: standard deviation.

* Constraint violation.
[a] In [20], it was reported 0.9316824.
[b] In [19], it was reported 0.96957758.
[c] In [12], it was reported 0.931682.
[d] In [17], it was reported 0.931676.
[e] In [22], it was reported 0.9316823879.
[f] In [18], it was reported 0.931682388.
[g] In [28], it was reported 0.93168230.

Table 9
Results for the series–parallel system.

Method	(n_1,n_2,n_3, n_4,n_5)	r_1	r_2	r_3	r_4	r_5	R_s	Slack (g_1)	Slack (g_2)	Slack (g_3)	MPI (%)	NFE	SD
HNN–SC [6]	(3,3,1,2,3)	0.83819295	0.85506525	0.87885933	0.91140223	0.85035522	0.99996875	53	$1.08092394 \times 10^{-5}$	7.1108488402	25.2770	–	–
GA [7]	(2,2,2,2,4)	0.785452	0.842998	0.885333	0.917958	0.870318	0.99997418	40	1.194440	1.609289	9.5626	20,000	–
ES–PSO [20]	(2,2,2,2,4)	0.819655	0.844975	0.895509	0.895509	0.868449	0.999976648998558[a]	40	$1.37197921 \times 10^{-4}$	1.6092889667	0.0002895	–	3.91E−07
GA–SRS [19]	(3,3,2,1,3)	0.824846726	0.842816570	0.908173083	0.898699000	0.865463014	0.999969638668474[b]	62	$6.38655843 \times 10^{-5}$	6.104140277	23.0898	20,000	1.60776E−06
INGHS [24]	(2,2,2,2,4)	0.8198118626	0.8449506842	0.8956701585	0.8952327069	0.868438057445	0.9999766489	40	5.305414×10^{-5}	1.6092889667	0.0007116	75,000	1.4188E−05
CS [12]	(2,2,2,2,4)	0.81992709	0.84526766	0.89549155	0.89544069	0.86831878	0.999976648897388[a]	40	$1.112960939 \times 10^{-5}$	1.609288966	0.0007228	10,000	
GA–PSO [29]	(2,2,2,2,4)	0.819640	0.845091	0.895482	0.895517	0.868430	0.999976649287932	40	$-4.8540641 \times 10^{-4}$	1.6092889667	*	–	–
IA [11]	(2,2,2,2,4)	0.819591561	0.844951068	0.895428548	0.895522339	0.868490229	0.999976649036520	40	$5.9845376654 \times 10^{-8}$	1.6092889667	0.0001269	20,000	3.0E−21
DE [30]	(2,2,2,2,4)	0.81965932	0.84498074	0.89550642	0.89550643	0.86844775	0.999976649066076[a]	40	$1.961642794 \times 10^{-7}$	1.609288966	4.1126×10^{-7}	32,000	–
TS–DE [31]	(2,2,2,2,4)	0.819659	0.844981	0.895507	0.895506	0.868448	0.9999766491	40	$-2.66935542 \times 10^{-4}$	1.609288966	*	–	–
ABC [8]	(2,2,2,2,4)	0.819737753469	0.844991099776	0.895529543820	0.895433687206	0.868434824469	0.999976649054	40	1.39152×10^{-10}	1.609288966	5.2126×10^{-5}	20,000	3.18206E−11
GA–PSO [46]	(4,3,2,3,2)	0.860328	0.827071	0.872994	0.937884	0.701148	0.999504300525587[c]	4	−22.2116202	−13.7167075	*	–	–
IAs [10]	(2,2,2,2,4)	0.812485	0.843155	0.897385	0.894516	0.870590	0.99997658	40	2.627×10^{-3}	1.609289	0.2949	13,500	–
LXPM–IPSO–GS [25]	(2,2,2,2,4)	0.819509	0.845312	0.895498	0.895568	0.868396	0.999976648938797[a]	40	$-1.718780531 \times 10^{-4}$	1.6092889667	*	–	–
AR–ICA [27]	(2,2,2,2,4)	0.82201264	0.84365640	0.89129092	0.89869886	0.86824939	0.99997661	40	3.96×10^{-4}	1.609289	0.1670	13,500	2.58E−06
IPSO [23]	(2,2,2,2,4)	0.81918526	0.84366421	0.89472992	0.89537628	0.86912724	0.99997664	40	5.61×10^{-4}	1.609289	0.03881	13,500	1.3362E−05
ABC [21]	(2,2,2,2,4)	0.8197457	0.8450080	0.8954581	0.9009032	0.8684069	0.99997731	40	−1.4695223388[a]	1.6092889667	*	–	–
ICS [14]	(2,2,2,2,4)	0.819927087	0.845267657	0.895491554	0.895440692	0.868318775	0.999976649	40	1.61×10^{-5}	1.6092890	0.0002833	10,000	4.45034E−06
CS [9]	(2,2,2,2,4)	0.819483232488	0.844783084455	0.895810553887	0.895220216915	0.868542486973	0.999976648818	40	$2.7216628 \times 10^{-10}$	1.609288966	0.001062	–	1.14187E−06
BBO [1]	(2,2,2,2,4)	0.8196583448	0.8449101406	0.8954871713	0.8955148963	0.8684681613	0.999976649048875	40	$1.748541669 \times 10^{-5}$	1.6092889667	7.4074×10^{-5}	20,000	0.11493E−09
MPSO [22]	(2,2,2,2,4)	0.8196547522	0.8449752789	0.8955087772	0.8955091117	0.8684491638	0.999976649066114[d]	40	$4.783657914 \times 10^{-9}$	1.6092889667	2.4818×10^{-7}	–	3.87E−08
CS–GA [18]	(2,2,2,2,4)	0.819660256	0.844981615	0.895519305	0.895492245	0.868447587	0.999976649065963[a]	40	$-2.7177208 \times 10^{-7}$	1.6092889667	*	–	–
PSO [28]	(4,3,2,1,2)	0.84025282	0.88865099	0.62375055	0.93984950	0.75158691	0.99985845[a]	68	0.916915088	4.017703641	83.5034	100,000	–
SSO [28]	(2,2,2,2,4)	0.81385803	0.83912659	0.89366150	0.89845276	0.87106323	0.99997657[a]	40	0.0024	1.609288966	0.3374	100,000	–
PSSO [28]	(2,2,2,2,4)	0.81958939	0.84458412	0.89534134	0.89581626	0.86852902	0.999976648738107[a]	40	$8.11794613 \times 10^{-5}$	1.609288966	0.001404	100,000	0.000232
NAFSA [38]	(2,2,2,2,4)	0.819787575273	0.845671943728	0.894868363315	0.895908268569	0.868295830551	0.999976648004	40	3.1248×10^{-8}	1.609289	0.004548	50,000	1.285E−05
PSFS (Proposed approach)	(2,2,2,2,4)	0.81965939118	0.84498085296	0.89550643076	0.89550645172	0.86844769346	**0.999976649066172**	40	$1.850821718 \times 10^{-10}$	1.6092889667	–	**5000**	**1.7E−25**

Slack: the unused resource.
NFE: number of function evaluations.
SD: standard deviation.

* Constraint violation.
[a] In [12,18,20,25,28,30], it was reported 0.99997665.
[b] In [19], it was reported 0.999988249.
[c] In [46], it was reported 0.99999988.
[d] In [1,22], it was reported 0.9999766491.

Table 10
Results for the complex (bridge) system.

Method	(n_1,n_2,n_3, n_4,n_5)	r_1	r_2	r_3	r_4	r_5	R_s	Slack (g_1)	Slack (g_2)	Slack (g_3)	MPI (%)	NFE	SD
HNN–SC [6]	(3,3,2,3,2)	0.81448276	0.82138254	0.89615152	0.71309103	0.81409107	0.99978937	27	7.0527393×10^{-6}	10.57247574	47.6036	–	–
GA [17]	(3,3,3,3,1)	0.808258	0.866742	0.861513	0.716608	0.766894	0.999889	18	9×10^{-4}	4.264770	0.5743	14,000	–
GA [7]	(3,3,3,3,1)	0.814090	0.864614	0.890291	0.701190	0.734731	0.99987916	18	0.376347	4.264770	8.6705	20,000	–
ES–PSO [20]	(3,3,2,4,1)	0.828082	0.857812	0.914241	0.648155	0.704066	0.999889638110263[a]	5	$-2.3814399 \times 10^{-4}$	1.560466288	*	–	–
GA–SRS [19]	(3,3,3,3,1)	0.80457234	0.85717305	0.86734683	0.72759162	0.76416666	0.999886726841704[b]	18	$6.77854912 \times 10^{-5}$	4.264769804	2.5696	20,000	4.47953E−07
INGHS [24]	(3,3,2,4,1)	0.8279847911	0.8576796813	0.9141564522	0.6484814055	0.7048654988	0.9998896364	5	1.8985×10^{-6}	1.560466288	0.001007	75,000	5.1860E−05
CS [12]	(3,3,2,4,1)	0.82809404	0.85800449	0.91416292	0.64790779	0.70456598	0.999889636778246[a]	5	$7.86629197 \times 10^{-5}$	1.560466288	0.0006649	10,000	
GA–PSO [29]	(3,3,2,4,1)	0.828134	0.857831	0.914192	0.648069	0.704476	0.999889638109295[a]	5	$-2.7258899 \times 10^{-4}$	1.560466288	*	–	–
IA [11]	(3,3,3,3,1)	0.816624176	0.868767396	0.858748781	0.710279379	0.753429200	0.9998893505	18	4.0420871×10^{-8}	4.264770	0.2593	20,000	4.0E−20
ABC [8]	(3,3,2,4,1)	0.827970276262	0.857874758586	0.914186404228	0.648355386813	0.703575311047	0.999889635809	5	$3.74636763 \times 10^{-4}$	1.560466288	0.001543	20,000	8.667E−09
GA–PSO [46]	(4,3,3,3,1)	0.858430	0.700000	0.922386	0.700000	0.700000	0.999263964662971[c]	11	−24.896621498	−27.3901210	*	–	–
IAs [10]	(3,3,3,3,1)	0.812485	0.867661	0.861221	0.713852	0.756699	0.99988921	18	1.494×10^{-3}	4.264770	0.3858	13,500	–
LXPM–IPSO–GS [25]	(3,3,3,3,1)	0.827974	0.857818	0.914166	0.648348	0.704427	0.999889636851887[a]	5	$3.35125258 \times 10^{-5}$	1.560466288	0.0005981	20,000	9.498E−06
AR–ICA [27]	(3,3,2,4,1)	0.82764257	0.85747845	0.91419677	0.64927379	0.70409200	0.99988963	5	4.428×10^{-5}	1.56046629	0.006806	13,500	1.0371E−04
IPSO [23]	(3,3,2,4,1)	0.82868361	0.85802567	0.91364616	0.64803407	0.70227595	0.99988963	5	3.59×10^{-6}	1.56046629	0.006806	13,500	4.0163E−05
DE [30]	(3,3,2,4,1)	0.82808641	0.85780478	0.91424067	0.64814622	0.70416210	0.999889637503023[a]	5	$1.987824296 \times 10^{-5}$	1.560466288	8.1949×10^{-6}	32,000	–
TS–DE [31]	(3,3, 2,4,1)	0.828086	0.857805	0.914241	0.648146	0.704162	0.999889637462077[d]	5	$3.71181366 \times 10^{-5}$	1.560466288	4.5296×10^{-5}	60,000	1.3290E−07
ABC [21]	(3,3,2,4,1)	0.828087	0.857805	0.704163	0.648146	0.914240	0.999484073426439[e]	5	−25.43392577	1.560466288	*	–	–
ICS [14]	(3,3,2,4,1)	0.828094038	0.858004485	0.914162924	0.647907792	0.704565982	0.999889636776613[a]	5	$7.93626251 \times 10^{-5}$	1.560466288	0.0006663	10,000	1.40652E−05
CS [9]	(3,3,2,4,1)	0.827855652338	0.857626105413	0.914752916604	0.648217208595	0.702670374782	0.999889631978	5	$1.06723518 \times 10^{-10}$	1.560466288	0.005014	–	7.037994E−07
NMDE [32]	(3,3,2,4,1)	0.8280864668	0.8578047868	0.9142407172	0.6481460609	0.7041626466	**0.999889637550196**[a]	5	$1.44763419 \times 10^{-8}$	1.560466288	– 3.4548×10^{-5}	40,000	–
BBO [1]	(3,3,2,4,1)	0.8280606892	0.8580404545	0.9141487486	0.6479689012	0.7042048796	0.9998896364	5	1.4541×10^{-4}	1.560466	0.001007	20,000	4.47686E−09
MPSO [22]	(3,3,2,4,1)	0.8280816704	0.8578118137	0.9142411461	0.6481547109	0.7040665038	0.999889637544786[d]	5	$-9.141729151 \times 10^{-9}$	1.560466288	*	–	–
CS–GA [18]	(3,3,2,4,1)	0.82808567	0.85780605	0.91424006	0.64814375	0.70418228	0.999889637551788[a]	5	$-7.3070123 \times 10^{-7}$	1.560466288	*	–	–
PSO–GC [13]	(3,3,2,4,1)	0.826678	0.857172	0.914629	0.648918	0.715290	0.99988957[a]	5	3.39×10^{-4}	1.560466	0.06113	13,500	6.9E−07
PSO [28]	(3,3,2,2,3)	0.77061588	0.90109253	0.89278651	0.60083008	0.73451002	0.99967140	37	16.54571312	1.411803224	66.4143	100,000	–
SSO [28]	(3,3,2,4,1)	0.82008362	0.85119629	0.91854858	0.66072083	0.70275879	0.99988862	5	0.00437599	1.560466288	0.913550	100,000	–
PSSO [28]	(3,3,2,4,1)	0.82783292	0.85771241	0.91437458	0.64861002	0.70287554	0.999889635738075[a]	5	$2.50290205 \times 10^{-5}$	1.560466288	0.001607	100,000	0.000502
NAFSA [38]	(3,3,2,4,1)	0.82832179189	0.85797450730	0.91422098825	0.64775717018	0.70300666185	0.99988963601	5	1.5485×10^{-5}	1.56046629	0.001361	50,000	2.1017E−05
PSFS (Proposed approach)	(3,3,2,4,1)	0.82812141729	0.85781341076	0.91423927822	0.64807680660	0.70424641245	0.999889637512067	5	$2.854643724 \times 10^{-6}$	1.560466288	–	**5000**	**3.7E−23**

Slack: the unused resource.
NFE: number of function evaluations.
SD: standard deviation.

* Constraint violation.
[a] In [12,14,18,20,25,28,29,30,32], it was reported 0.99988964.
[b] In [19], it was reported 0.99997413.
[c] In [21], it was reported 0.99988962.
[d] In [22,31], it was reported 0.9998896376.
[e] In [19], it was reported 0.99997413.

Table 11
Results for the overspeed protection system.

Method	(n_1,n_2,n_3,n_4,n_5)	r_1	r_2	r_3	r_4	R_s	Slack (g_1)	Slack (g_2)	Slack (g_3)	MPI (%)	NFE	SD
INGHS [24]	(5,5,4,6)	0.9015565830	0.8882438856	0.9481110971	0.8499817375	0.9999546745	55	5.05411×10^{-5}	24.8018827221	0.0003900	75,000	1.5221E−05
CS [12]	(5,5,4,6)	0.90161460	0.88822337	0.94814103	0.84992090	0.999954674680561[a]	55	$-1.04641326 \times 10^{-5}$	15.363463087	*	–	–
IA [11]	(5,5,4,6)	0.901588628	0.888192380	0.948166022	0.849969792	0.999954674554580[b]	55	$1.24953721 \times 10^{-4}$	15.363463087	0.0002696	18,000	4.14E−18
TS–DE [31]	(5,6,4,5)	0.901615	0.849921	0.948141	0.888223	0.999954674608111[c]	55	$1.89670533 \times 10^{-4}$	24.80188272	0.0001515	60,000	1.2504E−06
ABC [8]	(5,5,4,6)	0.901626809561	0.888208355883	0.948134377884	0.849942135673	0.999954674663	55	$5.57304247 \times 10^{-9}$	15.3634630874	3.0406×10^{-5}	20,000	3.38683E−11
IAs [10]	(5,5,5,5)	0.903800	0.874992	0.919898	0.890609	0.999942	50	2.152×10^{-3}	28.803701	21.8528	13,500	–
LXPM–IPSO–GS [25]	(5,5,4,6)	0.90163317	0.888251065	0.948141377	0.849854043	0.999954674599810[b]	55	$5.30549334 \times 10^{-6}$	15.36346308	0.0001698	20,000	–
AR–ICA [27]	(5,6,4,5)	0.90148988	0.85003526	0.94812952	0.88823833	0.999954673	55	2.13782×10^{-3}	24.8018827	0.003699	13,500	2.204E−05
IPSO [23]	(5,6,4,5)	0.90163164	0.84997020	0.94821828	0.88812885	0.999954674264402[d]	55	$9.45908976 \times 10^{-6}$	24.8018827	0.0009098	13,500	1.3895E−05
NMDE [32]	(5,6,4,5)	0.90161480	0.84992111	0.94814139	0.88822286	0.99995467	55	1.057×10^{-5}	24.80188272	0.0103	40,000	–
ABC [21]	(5,6,4,5)	0.901614	0.849920	0.948143	0.888223	0.999954674798458[d]	55	$-3.36378261 \times 10^{-4}$	24.80188272	*	–	–
DE [30]	(5,6,4,5)	0.90161482	0.84992114	0.94814139	0.88822284	0.99995467	55	$-1.00507377 \times 10^{-5}$	24.80188272	*	–	–
ICS [14]	(5,5,4,6)	0.901614595	0.888223369	0.948141029	0.849920899	0.999954674676423[a]	55	$9.643684961 \times 10^{-7}$	15.36346308	7.9215×10^{-7}	10,000	4.45034E−06
CS [9]	(5,5,4,6)	0.901598077027	0.888226184172	0.948101861662	0.849980778637	0.999954674585	55	$8.82494077 \times 10^{-10}$	15.3634630874	0.0002024	–	6.97619E−09
BBO [1]	(5,5,4,6)	0.9015629232	0.8882249447	0.9481559581	0.8499528951	0.9999546746	55	2.7021×10^{-5}	15.3634631	0.0001694	20,000	0.17816E−09
MPSO [22]	(5,6,4,5)	0.9016123483	0.8499199719	0.9481399512	0.8882260306	0.999954674676319[c]	55	$1.901702262 \times 10^{-7}$	24.8018827	1.0214×10^{-6}	–	4.23E−08
CS–GA [18]	(5,5,4,6)	0.901613407	0.888223375	0.948142110	0.849920787	0.999954674676407[b]	55	8.9299226×10^{-7}	15.36346308	8.2742×10^{-7}	30,000	3.11E−06
PSO–GC [13]	(5,6,4,5)	0.902231	0.856325	0.948145	0.883156	0.999953	55	0.975465	24.801882	3.5631	**2250**	1.1E−05
PSO [28]	(4,6,5,5)	0.92952331	0.81370356	0.88663747	0.89987183	0.99990474	37	11.5265677	11.6447077	52.4193	100,000	–
SSO [28]	(5,6,4,5)	0.90208435	0.85472107	0.94606018	0.88633728	0.99995416	55	0.109233104	24.8018827	1.1227	100,000	1.4805E−04
PSSO [28]	(5,5,4,6)	0.90166461	0.88817296	0.94821033	0.84987084	0.99995467	55	3.2872253×10^{-5}	15.36346308	0.01031	100,000	–
NAFSA [38]	(5,6,4,5)	0.90160779120	0.84993077684	0.94814603278	0.88821809379	0.99995467467	55	4.5195×10^{-7}	24.802	1.4963×10^{-5}	50,000	4.43E−06
PSFS (Proposed approach)	(5,6,4,5)	0.9016147331806	0.8499211645063	0.9481413898152	0.8882228732368	**0.999954674676782**	55	$5.3387338 \times 10^{-10}$	24.801882722	–	5000	**2.8E−20**

Slack: the unused resource.
NFE: number of function evaluations.
SD: standard deviation.

* Constraint violation.
[a] In [12,14], it was reported 0.99995468.
[b] In [11,18,25], it was reported 0.999954675.
[c] In [22,31], it was reported 0.9999546747.
[d] In [21,23], it was reported 0.999955.

Table 12
Results for the complex bridge network system.

Method	r_1	r_2	r_3	r_4	r_5	C_s	R_s	NFE	SD
ABC [8]	0.93545253	0.93438965	0.79041103	0.93541044	0.93448154	5.019918754645[a]	0.99	20,000	7.5105E−06
MRPS [36]	0.93479	0.93493	0.79219	0.93487	0.93496	5.01992	0.99	–	–
C–SOMGA [35]	0.935359	0.934304	0.790332	0.935504	0.934575	5.01992	0.99	100,000	–
I–NESA [34]	0.93747	0.93291	0.78485	0.93641	0.93342	5.01993	0.99	100,000	–
FGO [33]	0.93635	0.93869	0.80615	0.93512	0.93476	5.02042	0.9905	–	–
PSFS (Proposed approach)	0.93489568729	0.93489567889	0.79204120018	0.93489567384	0.93489573430	**5.019918127360**	0.99	**5000**	**1.6E−11**

NFE: number of function evaluations.
SD: standard deviation.
[a] In [8], it was reported 5.01991875.

Table 13
Results for the life-support system in a space capsule.

Method	r_1	r_2	r_3	r_4	C_s	R_s	NFE	SD
SA [34]	0.82529	0.89169	0.62161	0.72791	390.6327	0.990003	–	–
I–NESA [34]	0.825160	0.890130	0.62825	0.72917	390.572	0.99000	100,000	–
FGO [33]	0.82569	0.89022	0.62732	0.72865	390.558646616[a]	0.989999013262[*]	–	–
PSFS (Proposed approach)	0.825582317423114	0.890154652678330	0.627662618976073	0.728849479495775	**390.5707384241**	0.990000000	**5000**	**7.3E−10**

NFE: number of function evaluations.
SD: standard deviation.
[*] Constraint violation.
[a] In [33], it was reported 390.570.

Table 14
Results for the large scale reliability–redundancy allocation problem.

Method	$(n_1,n_2,n_3,n_4,n_5,n_6,n_7,n_8,n_9,n_{10},n_{11},n_{12},n_{13},n_{14},n_{15},n_{16},n_{17},n_{18},n_{19},n_{20})$	$(r_1,r_2,r_3,r_4,r_5,r_6,r_7,r_8,r_9,r_{10},r_{11},r_{12},r_{13},r_{14},r_{15},r_{16},r_{17},r_{18},r_{19},r_{20})$	R_s	Slack (g_1)	Slack (g_2)	Slack (g_3)	MPI (%)	NFE	SD
IPSO [25]	–	–	0.884232577	–	–	–	5.428756	40,000	0.0227
HS–IPSO [25]	–	–	0.886724346	–	–	–	3.348436	40,000	0.0188
LXPM–IPSO [25]	–	–	0.887779879	–	–	–	2.439339	40,000	0.0149
HS–IPSO–GA [25]	–	–	0.888685306	–	–	–	1.645787	40,000	0.0201
PSFS (Proposed approach)	(2,2,3,2,3,2,3,3,3,3,3,3,3,3,3,3,3,3,3,2)	(0.920682125899, 0.952579760087, 0.840370879766, 0.934499487329, 0.806884188682, 0.895206390582, 0.811801524606, 0.814181963158, 0.836220223575, 0.827983638973, 0.814585208335, 0.837346324449, 0.841065088128, 0.821075460589, 0.850117124276, 0.838491056144, 0.823073269011, 0.809956458845, 0.807719250830, 0.897948276899)	**0.89051730902**	143	0.00275018	2.1970797	–	**5000**	**3.5E−08**

NFE: number of function evaluations.
SD: standard deviation.

confirms that the solutions of the PSFS are better than all the other methods considered in terms of maximizing the system reliability for the problem of Table 11 and minimizing the cost for the problem of Tables 12 and 13. Furthermore, the PSFS showed more stability with low NFE.

Table 14 compares the results obtained by the PSFS for the large scale reliability–redundancy allocation problem with those of other works available in the literature. It can be seen that the overall reliability provided by the present approach (R_s=0.890517) outperforms the IPSO [25] (R_s=0.884232), HS–IPSO [25]

Table 15
Results for the 10-unit structure system.

Scenario	Method	$(n_1,n_2,n_3,n_4,n_5,n_6,n_7,n_8,n_9,n_{10})$	R_s	Slack (g_y)	MPI (%)	NFE	SD
10 × 1	GA [12]	–	0.57473724	–	100	100,000	1.1723E−01
	ABC [12]	–	0.95403857	–	99.999909	20,000	9.9360E−02
	HS [12]	–	0.88752850	–	99.999963	25,000	1.9472E−01
	NGHS [12]	–	0.98201395	–	99.999769	25,000	9.4964E−03
	CS [12]	–	0.98335697	–	99.999750	7500	5.5595E−04
	AGGA [37]	–	0.852953	–	99.999971	–	–
	ASACO [37]	–	0.999992	–	99.480835	5625	–
	PSFS (Proposed approach)	(15,13,15,35,1,1,1,1,1,1)	**0.999999958466866**	0.353894500	–	**5000**	**6.4E−09**
10 × 5	GA [12]	–	0.40859193	–	99.783885	100,000	9.9898E−02
	ABC [12]	–	0.63631535	–	99.648563	20,000	8.8019E−02
	HS [12]	–	0.50157124	–	99.743569	25,000	9.3011E−02
	NGHS [12]	–	0.64820235	–	99.636688	25,000	4.4999E−02
	CS [12]	–	0.67189992	–	99.610447	7500	3.3645E−16
	AGGA [37]	–	0.817292	–	99.300456	–	–
	PSFS (Proposed approach)	(4,4,5,6,1,1,2,3,2,2)	**0.998721878906043**	(441.1035, 365.2964, 44.3940, 4.2572)	–	**5000**	**7.2E−19**

NFE: number of function evaluations.
SD: standard deviation.

Table 16
Results for the 15-unit structure system.

Scenario	Method	$(n_1,n_2,n_3,n_4,n_5,n_6,n_7,n_8,n_9,n_{10},n_{11},n_{12},n_{13},n_{14},n_{15})$	R_s	Slack (g_y)	MPI (%)	NFE	SD
15 × 1	GA [12]	–	0.82700343	–	99.998214	100,000	1.1514E−01
	ABC [12]	–	0.95260704	–	99.993484	20,000	3.4596E−03
	HS [12]	–	0.97235030	–	99.988813	25,000	7.0892E−04
	NGHS [12]	–	0.97235030	–	99.988813	25,000	9.9082E−04
	CS [12]	–	0.97276197	–	99.988662	7500	3.6879E−04
	AGGA [37]	–	0.914283	–	99.996397	–	–
	ASACO [37]	–	0.999993	–	55.884162	5625	–
	PSO [9]	–	0.97967292	–	99.984807	–	4.091E−03
	CS [9]	–	0.98421284	–	99.980439	–	0.003477
	ICS [14]	–	0.97235030		99.988813	5000	6.4903E−05
	PSFS (Proposed approach)	(12,10,14,30,9,8,1,1,1,1,1,1,1,1,1)	**0.999996911891346**	0.2567339	–	5000	**8.1E−09**
15 × 5	GA [12]	–	0.82381093	–	99.579818	100,000	1.3423E−01
	ABC [12]	–	0.95141221	–	98.476337	20,000	2.9954E−03
	HS [12]	–	0.95825279	–	98.226674	25,000	1.0764E−03
	NGHS [12]	–	0.95810719	–	98.232837	25,000	1.8335E−03
	CS [12]	–	0.95825279	–	98.226674	7500	2.2199E−04
	PSO [9]	–	0.97390351	–	97.163166	–	5.614E−03
	CS [9]	–	0.97503662	–	97.034399	–	0.003785
	ICS [14]	–	0.95825279	–	98.226674	5000	7.1692E−05
	PSFS (Proposed approach)	(8,5,7,9,6,7,1,1,3,2,1,1,1,1,1)	**0.999259685931657**	(471.6977, 609.3498, 5.5301, 53.2939, 402.9430)	–	5000	**1.4E−09**

NFE: number of function evaluations.
SD: standard deviation.

(R_s=0.886724), LXPM–IPSO [25] (R_s=0.887779), and HS–IPSO–GA [25] (R_s=0.888685). The problem involves forty design variables, including twenty integers. The improvements made in the present paper over the other works are 5.428756%, 3.348436%, 2.439339%, and 1.645787%, respectively. All the previous approaches used 40,000 function evaluations and their respective SD were 0.0227, 0.0188, 0.0149 and 0.0201, whereas the present approach used 5,000 function evaluations and the SD is 3.5E−08.

From Tables 15 and 16, it can be observed that the best solutions are given by the implemented PSFS for the 10 and 15 unit structure systems by considering the scenarios (10 × 1; 10 × 5) and (15 × 1; 15 × 5), respectively. The numbers of redundancies have not been provided by the works available in the literature. Thus, the constraint violations cannot be verified.

Finally, the results for the reliability–redundancy allocation problem of the pharmaceutical plant are reported in Table 17. The maximum system reliability obtained by the PSFS is 0.962291, whereas in [16] the author found 0.956021 using the PSO. Therefore, the implemented PSFS improved the value of R_s over the PSO by 14.258672%. From Tables 15–17, it can be observed that the best system reliability is obtained by the proposed approach with more stability and consuming lower NFE. From all the previous case studies, it can be observed that the challenge in solving the ROPs is to maximize the overall system reliability within the resource limits whatever the configurations of the subsystems and the nature of the design variables (real, integer, or mixed).

7. Conclusions

The goal of this paper was to implement an efficient approach for solving reliability optimization problems. A penalty guided

Table 17
Results for the pharmaceutical plant.

Parameters	Optimal values	
	PSO [16]	**PSFS (Proposed approach)**
(r_1, n_1)	(0.871922, 3)	(0.88380942954, 3)
(r_2, n_2)	(0.827480, 3)	(0.82401235102, 3)
(r_3, n_3)	(0.835569, 3)	(0.82854756196, 3)
(r_4, n_4)	(0.800000, 3)	(0.82778491734, 3)
(r_5, n_5)	(0.865663, 3)	(0.86606546371, 3)
(r_6, n_6)	(0.831345, 3)	(0.83549324163, 3)
(r_7, n_7)	(0.864687, 3)	(0.84838122612, 3)
(r_8, n_8)	(0.800000, 3)	(0.77741531196, 4)
(r_9, n_9)	(0.858897, 3)	(0.84940798704, 3)
(r_{10}, n_{10})	(0.832932, 3)	(0.8292440869, 3)
R_s	0.956021	**0.9622918216**
MPI (%)	14.258672	–
Slack (g_1)	0.053205	7.1536172×10^{-6}
Slack (g_2)	13.025996	15.8193601
Slack (g_3)	10.00000	3.00000
NFE	50,000	**5000**
SD	–	**2.8E − 12**

NFE: number of function evaluations.
SD: standard deviation.

stochastic fractal search has been presented and applied to ten numerical case studies, including redundancy allocation, reliability allocation and reliability–redundancy allocation problems. The results reveal that the proposed approach has been successfully implemented. Furthermore, the results of several other methods applied in the literature have been compared to those obtained in the present work and it has been proven the superiority of the proposed approach. Moreover, taking into account the criterions of small standard deviation and lower number of function evaluations clearly shows the better performance and robustness of the present approach for solving the system reliability optimization problem.

Since the present solution method is limited to a single objective function, future work will be devoted to the development of a hybrid PSFS for solving multiobjective reliability optimization problems.

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